NET ZERO CASE STUDIES REPORT:

DETAILED STUDIES OF BROCK ENVIRONMENTAL CENTRE AND PIKES PEAK SUMMIT VISITOR CENTRE





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Abstract

Net zero carbon discusses balancing carbon emissions, both operational and embodied carbon through renewables on-site, off-site or the purchase of carbon offsets. It requires deep incorporation of passive strategies, material selection and energy efficient active systems in buildings.

The Brock Environmental Centre (BEC) is a unique building owned by the Chesapeake Bay Foundation (CBF) located at Virginia. It is popular due to its net-zero energy, water and waste. The building produces eighty percent (80%) more energy than it consumes. The outstanding renewable energy features are solar photovoltaics, wind turbines and rainwater harvesting systems.

Pikes Peak Summit Visitor Centre is a pre-cast concrete and stone building located at the summit of one of the highest mountains in United States with very harsh conditions of winds, snow and storms at a high altitude. The major form of renewable energy that powers the building is a mega solar project within the state of Colorado.

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CASE STUDY 1: BROCK ENVIRONMENTAL CENTRE

1.0 INTRODUCTION

The single storey educational facility is a net zero Certified Living Building, and a Leadership in Energy and Environmental Design (LEED) platinum project situated along Virginia Beach in the United States of America. The centre accommodates offices, meeting rooms and a conference room. The main target of the project was net zero energy, water and climate resilience (Grable, 2017).

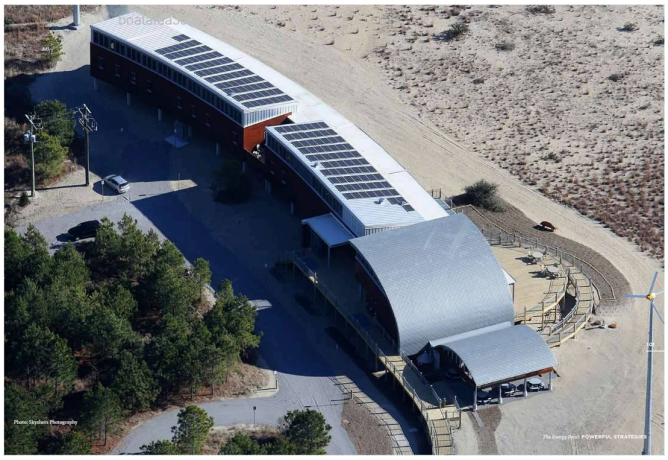


Figure 1: Aerial view of Brock Environmental Centre (Grable, 2017 p.100)

1.1 Project Location and Climatic Data

The project is situated at Pleasure House Point, Virginia at coordinates 36.8° N and 75.9° W (UCLA Energy Design Group, 2021). According to the Koppen-Geiger climate zones, the site lies within a humid sub-tropical zone (Beck *et al.*, 2018). Location has an average temperature of 18 degrees Celsius as seen in Figures 2 and 3 with lowest and highest temperatures of -10 and 33 degrees respectively. Relative humidity is high during night-time at 81.9% and average wind speed of 8m/s (Nash, 2014; UCLA Energy Design Group, 2021).

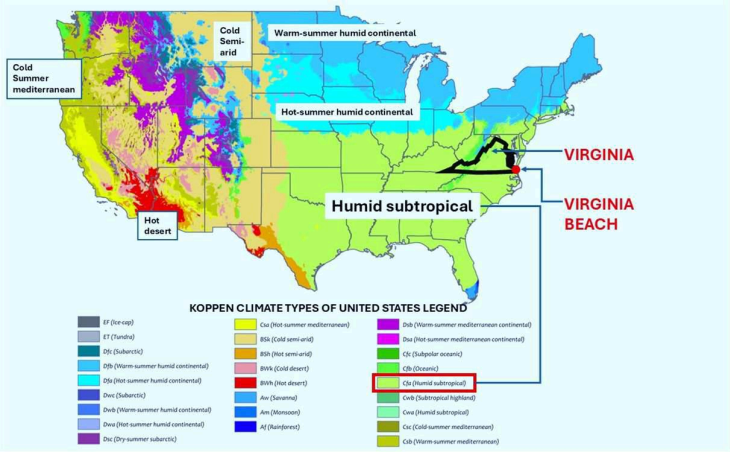


Figure 2: Köppen–Geiger climate zones (modified by author from; Beck et al.,2023 p.3)

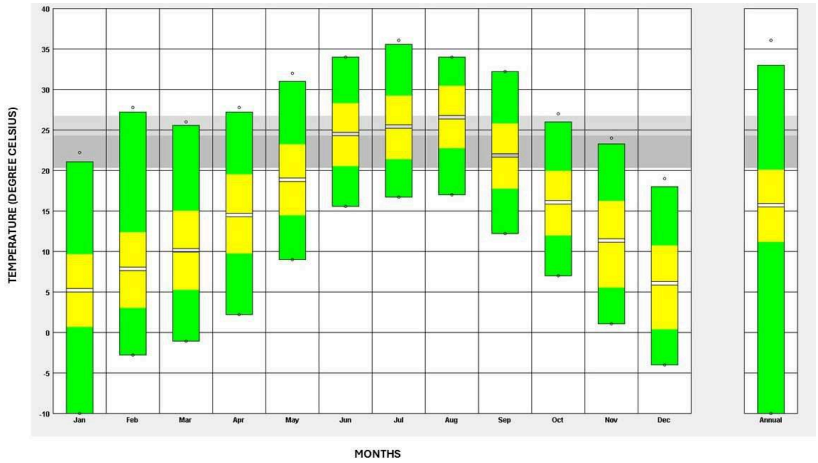


Figure 3: Temperature recordings of Virginia Beach (UCLA Energy Design Group, 2021)

1.2 Microclimate of Virginia Beach

Even though the entire area lies in a sub-humid terrain, the microclimate of the area is a mixture of sub-humid and oceanic due to its location along the Atlantic Ocean (U.S. Environmental Protection Agency, 2006). The area receives 46.3 inches of rain per year (Allen and Allen, 2019).

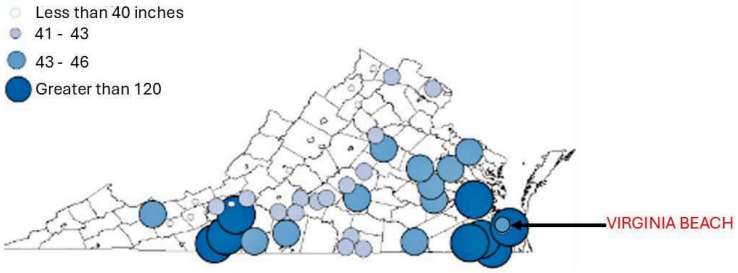


Figure 4: Mean total precipitation of Virginia (Allen and Allen, 2019, p. 11)

1.3 Net Zero Carbon Policies of United States

The United States (U.S.) has set a goal to limit global warming and reduce greenhouse gas emissions by 50-52% by 2030 as seen in Figure 5. The outstanding Nationally Determined Contributions (NDC) is; "*the goal for 100% carbon pollution-free electricity by 2035 and the goal for net-zero emissions no later than 2050*" (United States Department of State and the United States Executive Office of the President, 2021,p.10). Buildings will also receive hundred percent clean power, carbon-free electrification, improved energy efficiency, retrofitting and energy codes to support the transition to a low-carbon economy. (United States Department of State and the United States Executive Office of the President, 2021).

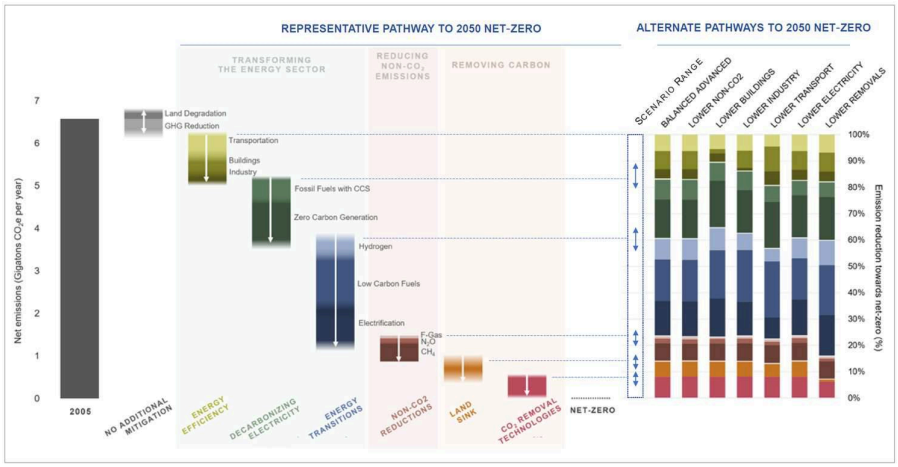


Figure 5: United States net zero carbon policies (United States Department of State and the United States Executive Office of the President, 2021, p. 23)

1.4 Project Details

Table 1: Project Details

Total Land Area	40460 m^2
Total Built-Up Area (5%)	975 m^2
Client	Chesapeake Bay Foundation (CBF)
Architect, Interior Designer and MEP Engineer	SmithGroup
Structural Engineer	A&F Engineers
Contractor	Hourigan Construction
Civil Engineer and Landscape Architect	WPL Site Design
LEED & Living Building Challenge Management	Janet Harrison Architect
Start Of Construction	21st August, 2013
Official Opening	November 14, 2014
Start Of Occupancy	5th January, 2015
Start Of Performance Evaluation	1st April, 2015
Certifications	LEED NC Platinum (2015) Living Building Challenge version 2.0 (2016)

Source: International Living Future Institute (n.d).

2.0 PASSIVE DESIGN STRATEGIES

2.1 Building site and orientation

Grable (2017) informs the site is located at a flood prone area and experiences extreme hurricanes that informed the building's location to be set 200 feet away from the shoreline. The centre has its' longest side facing south to maximize daylighting into interior spaces and increase passive heat gain as seen in Figure 6.

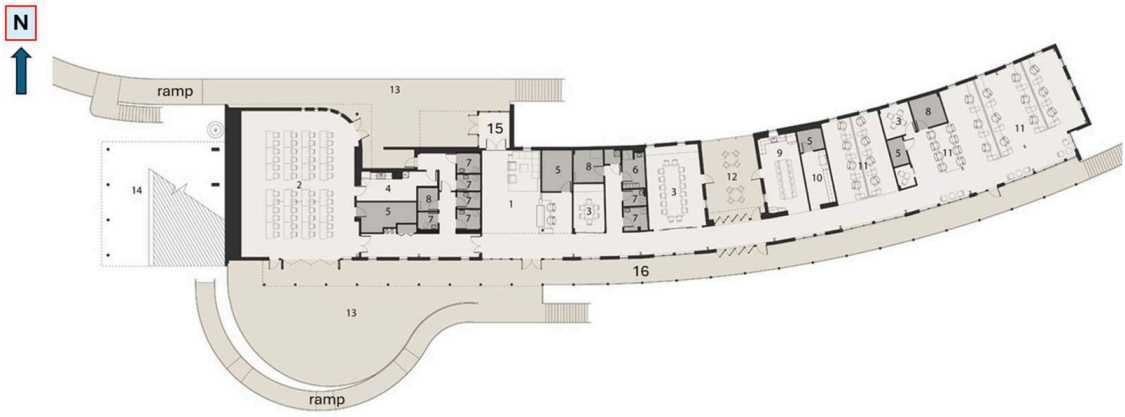


Figure. 6 Floor plan of Brock Environmental Centre (Grable, 2017 p.45)

Table 2: Designation of spaces of floor plan

1	Lobby	9	Staff dining room
2	Conference room	10	Copy room
3	Meeting room	11	Offices
4	Catering kitchen	12	Dog trot
5	Storage	13	Deck/porch
6	Staff shower	14	Education pavilion (below)
7	W/c	15	Draught lobby
8	Mechanical room	16	Long porch

Source: Grable (2017)

2.2 Building form and energy implications

The single storey building's form was influenced by native vernacular Virginia long houses (Grable, 2017). Modification of the original form increased the energy performance and solar heat gain especially from the south-east to capture the morning sun (SmithGroup,2015).

2.3 Shading

The building is shaded with awnings and overhangs. A long porch runs along the south façade to allow the sun to warm the south and prevent sunlight directly hitting the interior. (Grable, 2017) as seen in Figure 6.

2.4 Buffer zones

The draught lobby connects the ramp and stairs to the reception. The dogtrot (breezeway native to vernacular buildings of Virginia) improves ventilation and connects the interior spaces to the outdoor as highlighted by Grable (2017) in Figure 6.

2.5 Natural ventilation and windows

The building has a window-to-wall ratio of 20-30% which informed the window sizes. Stack and cross ventilation were efficiently employed in the building. The southwest breeze travels

through the dogtrot windows to the north clerestory, whiles the northeast breeze travels through the north windows to the south clerestory as depicted in Figure 7 (Grable, 2017).

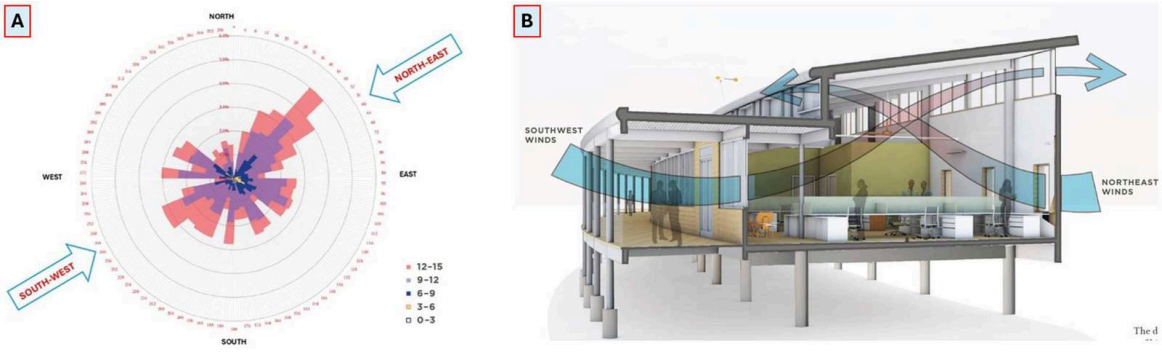


Figure 7: (A)Wind wheel (Grable, 2017 p.92) (B)Cross and stack ventilation (Coffield, Lappano and Greg, 2016 p.28)

2.6 Daylighting

Daylighting was a major strategy, allowing natural light into interior spaces. The south-facing porch serves as a buffer for direct sunlight. Daylight is diffused into office spaces though the clerestories (Grable, 2017).

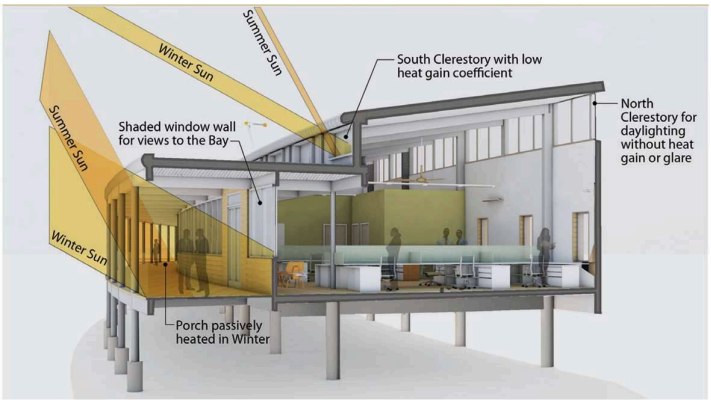


Figure 8: Daylighting strategies (Coffield, Lappano and Greg, 2016, p.30)

2.7 Building Resilience

Considering the flood prone and hurricane conditions of the site, meticulous steps were taken by the design team to mitigate future risks as seen on Table 4. In the year 2015, the building withstood hurricane Joaquin with strong winds and floodwaters as evidenced in Figure 9 (Urban Land Institute, n.d.).

Table 3: Resilience Strategies

Location challenge	**Strategy adopted on site and building**
Flooding from Atlantic Ocean	-set back the building 200 feet from the shoreline -elevate building 14 feet above sea level on concrete columns

Storm water	-Permeable road with gravels and pavement walkways -Rain gardens on various parts of the site -Water infiltration area planted with Juncus grass with fibrous, root systems that hold the soil
High winds and hurricane	Building's steel structure to withstand 120 m/hr winds
Emergency shutdown of power	Emergency battery backup

Source: SmithGroupJJR (2016)

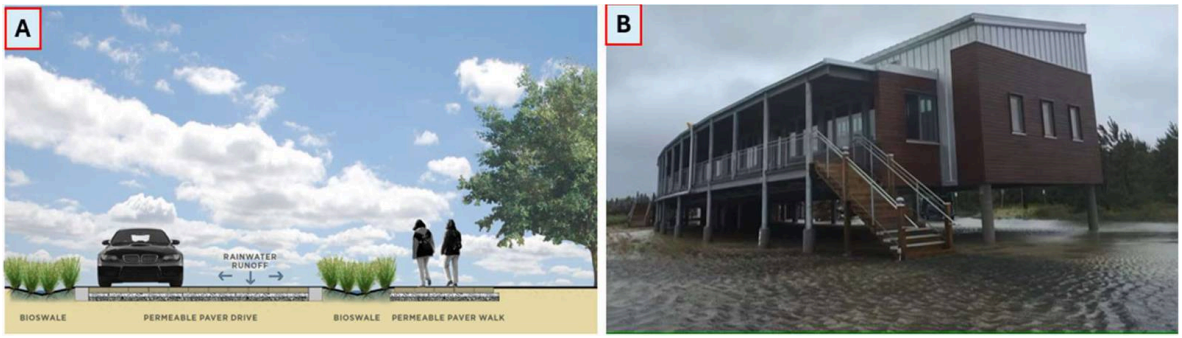


Figure 9: (A) Permeable parking and walkway (Grable, 2017 p.79) (B)Flooding at Brock Centre after Hurricane Joaquin in October 2015 (Chesapeake Bay Foundation, 2017)

2.8 Insulation and Air Tightness

Lappano (2015) states that designers deliberately incorporated passive solar design strategies for the winter sun to enter the interior spaces. The centre is excellently insulated with high R-values for all components as represented on Table 3.

Table 4: Envelope, components and R-Value

ELEMENT	COMPONENT(S)	R-VALUE
Roof - Galvalume	• Vented composite panels • Half inch plywood • One inch air space • 81/2 inches polyisocyanurate insulation • Vapour-permeable self-adhering underlayment • Standing seam galvalume	50
Exterior Walls	• 2×6 wood framing • Cavity filling of 5 ½ inches of high-density batt insulation • Exterior plywood sheathing • 3 inches of rigid extruded polystyrene (XPS) • Rainscreen assembly (furring strips of reclaimed cypress wood)	35
Floor Slab	• Concrete floor with insulation and wooden floor finish	R-31

Basement/Foundation	• Concrete	R-24.59
Windows (20-30 % wall-to-window ratio)	• Triple-paned • Argon filled • Solar heat gain coefficient 0.26	R-6.67

Source: (Chesapeake Bay Foundation, 2014; Grable, 2017)

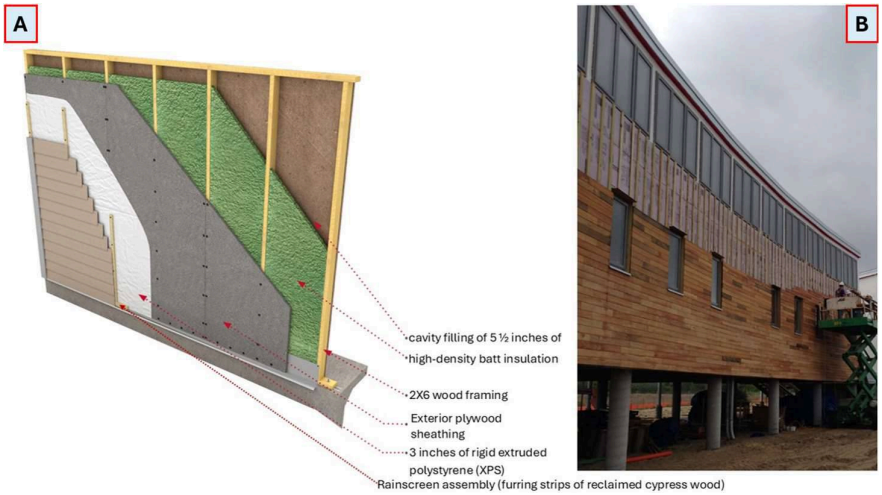


Figure 10: (A)Wall detail (Goodman, 2015) (B) Exterior façade showing rainscreen assembly (Grable 2017 p.133)

2.9 Envelope, Materials and Energy Implications

The foremost consideration for material selection was the exclusion of all red list materials and inclusion of non-toxic, biobased, recycled and salvaged materials from different parts of the region as summarised on Table 5. Full vetting of all materials was according to the Health Product Declaration (HPD) of United States (SmithGroupJJR, 2015).

Table 5: Salvaged Materials used for Project

SALVAGED MATERIALS	**SOURCE**
Solid wood interior doors	Local office demolition
Wood flooring	Reclaimed fence posts Salvaged gym flooring from local school
Granite countertops	Salvaged office credenza
Mirrors and toilet accessories	Local hospital
Lockers	Local high school
Ceramic tile	Overstock left over from previous project
Exterior rainwater cistern	Salvaged pickle barrels
Cypress siding	Reclaimed sinker logs
Lavatories and mop sinks	Local hospital
Reception desk millwork	Salvaged pickle barrel
Waste concrete as crushed aggregate	Different construction sites
Recycled foam (FOAMULAR)	FOAMULAR manufacturing company

Source: (Chesapeake Bay Foundation, 2014; Grable, 2017)

3.0 BUILDING SERVICES AND ENERGY

3.1 Lighting

Low mercury fluorescent pendant light emitting diode (LED) downlights and task lighting were the artificial lights specified which incorporated occupancy sensors that automatically turned on in the presence of occupants (International living future institute,n.d.).

3.2 Heating, Ventilation and Air Conditioning

Cool air is supplied to the building with a dedicated outside air system (DOAS) (Coffield, Lappano and Greg, 2016) as presented in Figure 11. Carbon dioxide sensors were incorporated with this system to regulate the supply of air to the spaces. A central water-cooled variable refrigerant flow (VRF) system was incorporated into the geothermal loop, presenting five percent (5 %) energy savings, SmithGroupJJR (2015).

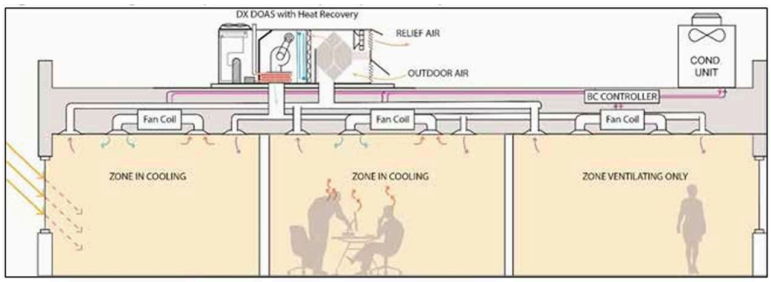


Figure 11: Dedicated Outside Air System and Variable Refrigerant Flow system (Northwest Energy Efficiency Alliance,2021 p.12)

3.3 Net Zero Water – Rainwater Harvesting and Water Treatment

According to Grabble (2017), the BEC operates fully as an office but uses 90 percent less water than the average office building. The centre harvests rainwater from the 510 msq roof, which travels through downspouts to a vortex filter and channelled into two (2) underground 1,650-gallon storage tanks. The harvested water goes through an ultraviolet chlorine treatment and the water is finally pumped into a clean tank with calcium carbonate and distributed to the building as seen in Figure 12.



Figure 12: Water harvesting and treatment process (Grable, 2017 p.)

4.0 NET ZERO ENERGY STRATEGIES AND RENEWABLE ENERGY

The centre exclusively produces electricity via solar (168 panels) and wind and produces 83 % more energy than it consumes. The photovoltaic panels (46.5 kW) fixed on the roof produces 60 % of the building's energy whiles the seventy (70) feet tall two (2) 10-kilowatt wind turbines produce 40 % of the building's energy as presented in Figures 13 and 14 (Lappano, 2015). The geothermal system consists of eighteen (18) vertical closed loop and 250 ft deep wells employed for ventilation, heating and cooling (Chesapeake Bay Foundation, n.d.).

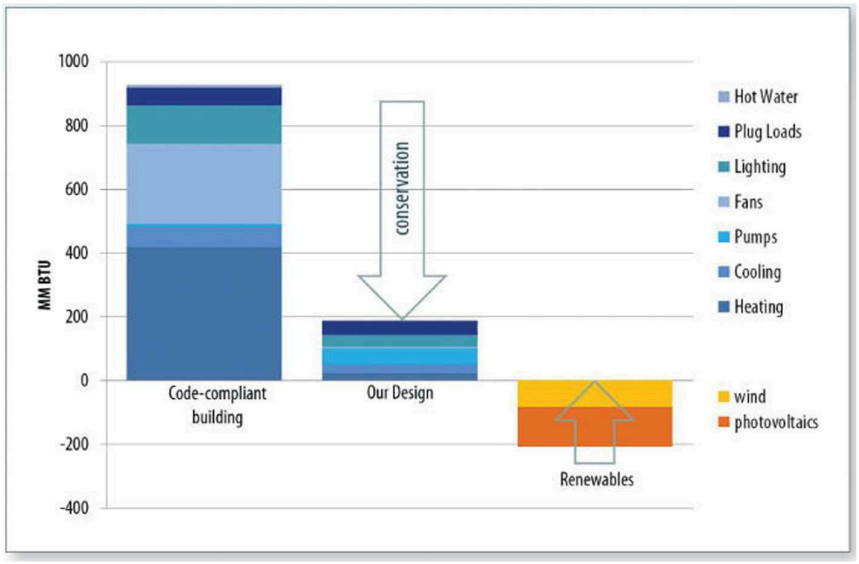


Figure 13: Reducing power consumption and incorporation of renewables (Lappano, 2015 p.30)

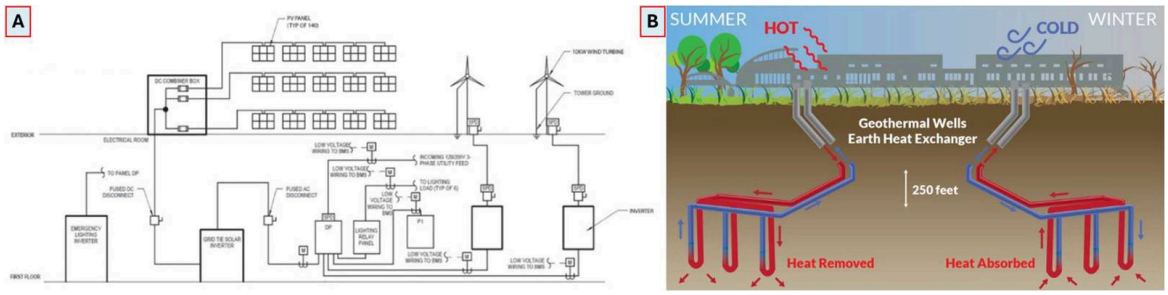


Figure 14: (A) Solar and wind renewable diagram (Lappano, 2015 p.31) (B) Geothermal heating and cooling (Chesapeake Bay Foundation, n.d.)

4.1 Building Performance (Predicted Versus Operational)

Due to the building's low water demand, less energy was required to pump and treat water. The initial Energy Use Intensity (EUI) prediction of the energy model was 20 kBtu/f^2 but was reduced to 16.5 kBtu/f^2, after critical design analysis. During the first year of its operation, the centre's EUI was a remarkable 14.14 kBtu/f^2 which accounted for 80 % energy savings

(Grable, 2017). Figure 16 and 17 present the actual performance and comparison with an average building's energy consumption.

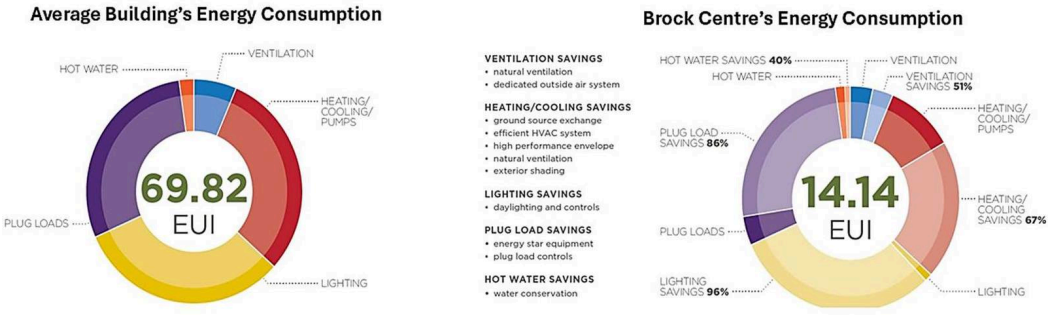


Figure 15: Comparison of an average building and Brock centre's energy consumption (Grable, 2017 p.170)

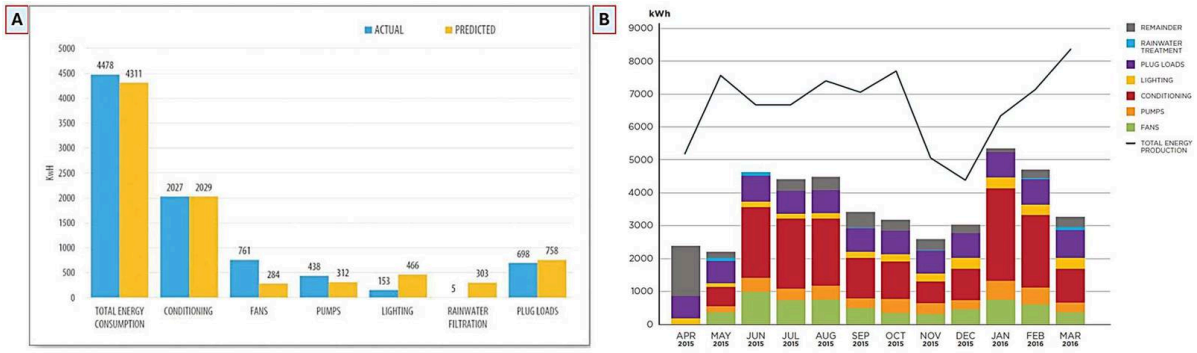


Figure 16: (A) Actual energy versus predicted energy consumption (Lappano, 2015 p.32) (B) Energy use versus total energy production (Coffield, Lappano and Greg, 2016)

5.0 EMBODIED CARBON

To achieve net zero carbon, passive design strategies were employed to drastically reduce energy, on-site renewable energy choices of solar photovoltaics, wind power and ground source heat pump were incorporated. Local and recycled materials were sourced within 500-km radius of the area as major construction materials (Ahuja and Chopson 2024).

Table 6: Strategies to reduce carbon emissions

Systems	Avoidance of carbon emitting systems – use of DOAS with VRF
Materials	-Local materials -Reuse of salvaged materials and construction waste -Elimination of red list materials -Incorporation of biobased materials
Carbon offset purchase	Purchase of 1,234 metric tonnes of carbon offsets to support a landfill gas to energy project for use as fuel
Transportation and parking	Parking offsite (during construction) for all vehicles except disabled parking and construction delivery trucks

During construction	- Mobile photovoltaic arrays that powered construction equipment - Exterior site lights installed with motion sensors to save energy - Generators, forklifts and excavators fitted with catalytic converters, to convert toxic gas to less toxic pollutants - Wells sunk to supply water for the entire construction. - Site office totally built using reclaimed materials, fitted with solar photovoltaic panels and a high-efficiency HVAC system.

Source: Grable (2017)

6.0 CERTIFICATIONS

LEED Platinum certification was awarded to the Brock Environmental Centre in 2015 after scoring more than 80 points during its evaluation based on the LEED criteria. The centre's performance exceeded the expectation of the Living Building Challenge (LBC) and was awarded the LBC 2.0 certificate in 2016 (SmithGroup, 2015).

7.0 POST OCCUPANCY EVALUATION

Grable (2017) informs that during the building performance audit year, a post-occupancy evaluation of the centre conducted using the Performance Measurements Protocol (PMP) based on six main areas: energy use, water, thermal comfort, indoor air quality, lighting and acoustics. All areas were satisfactory except acoustics due to external noise that came in through open windows as can be seen in Figure 17.

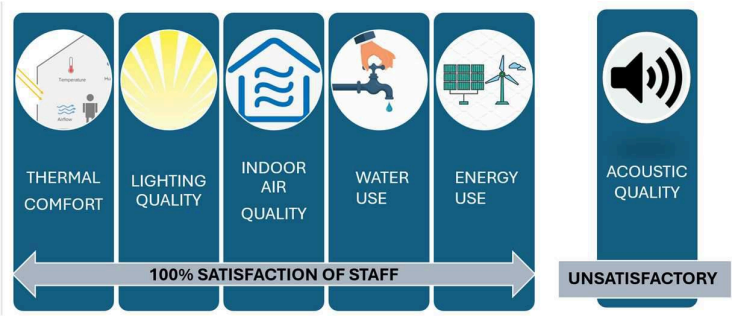


Figure 17: Staff satisfaction levels of Brock centre (Grable, 2017 p.170)

CASE STUDY 2: PIKES PEAK SUMMIT VISITOR CENTRE

1.0 INTRODUCTION

The Pikes Peak summit visitor centre is “an extraordinary project with extraordinary challenges” as stated by Grable (2024 p.4). It serves travellers, explorers and other guests who want to have a unique mountain experience. According to Huso (2024), this visitor centre is the first of a kind located at such a high altitude.

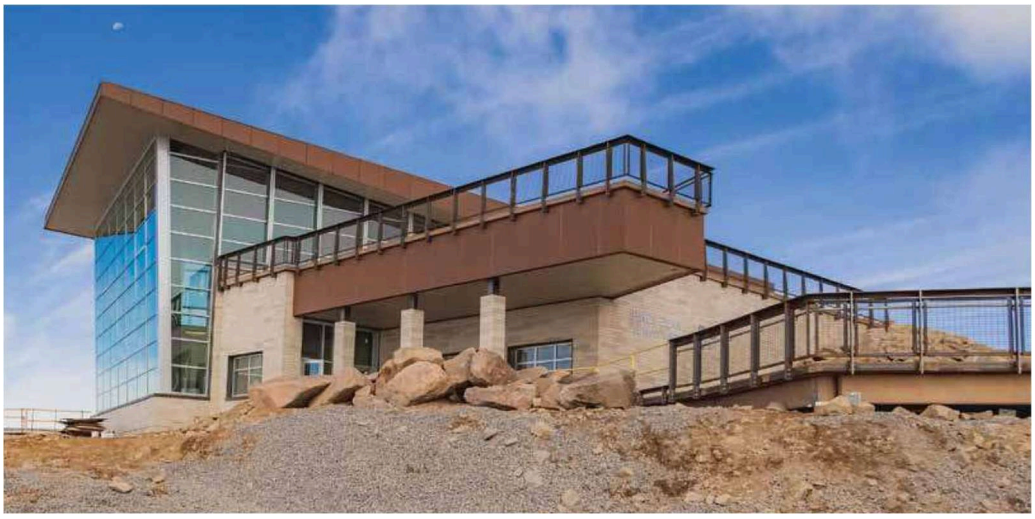


Figure 1: Southeast view of the Pikes Peak summit building (Huso, 2024 p.35)

1.2 Project Location and Climatic Data

Located on the summit of Pikes Peak Mountain in Colorado, the project is 4,302 m above sea level embracing the extreme harsh conditions of wind, snow, and thin air (Grable, 204). According to the Koppen-Geiger climate zones, the site lies within a tundra and sub-artic zone (Beck et al., 2018). The location has an average temperature of 11 degrees with extreme lowest and highest temperatures of -40 degrees and 23 degrees respectively and 230 m/h high winds (Meteoblue,2024 and Huso, 2024).

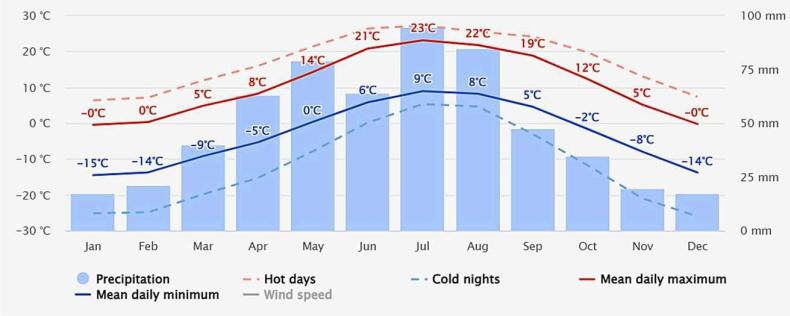


Figure 2: Temperature range of Pikes Peak Mountain Summit (Meteoblue, 2024)

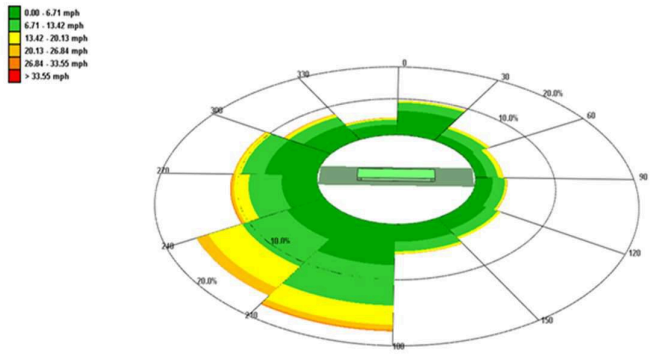


Figure 3: Wind speed of Pikes Peak Mountain (City of Colorado Springs, 2016, p.25)

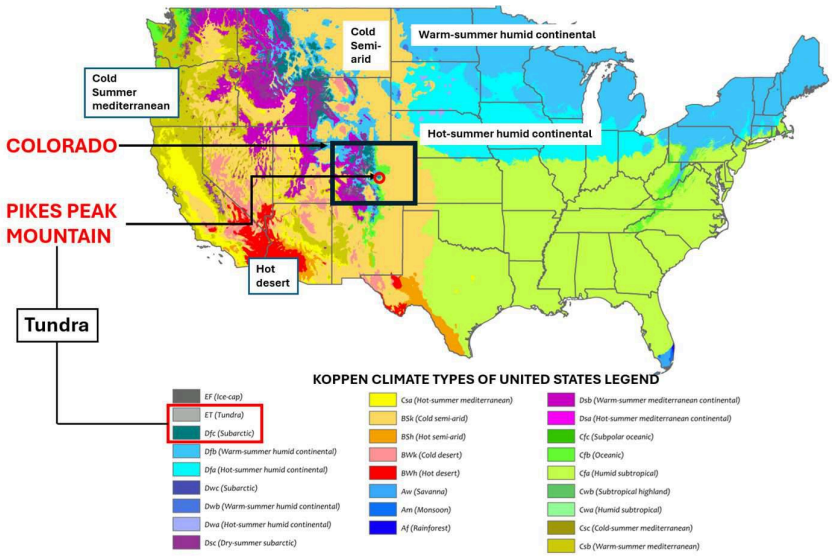


Figure 4: Köppen–Geiger climate zones (modified by author from; Beck et al.,2023 p.3)

1.3 Net Zero Carbon Policies of United States

The United States (U.S.) has set a goal to limit global warming and reduce greenhouse gas emissions by 50-52% by 2030 as seen in Figure 5. The outstanding Nationally Determined Contributions (NDC) is; "*the goal for 100% carbon pollution-free electricity by 2035 and the goal for net-zero emissions no later than 2050*" (United States Department of State and the United States Executive Office of the President, 2021,p.10). Buildings will also receive hundred percent clean power, carbon-free electrification, improved energy efficiency, retrofitting and energy codes to support the transition to a low-carbon economy. (United States Department of State and the United States Executive Office of the President, 2021).

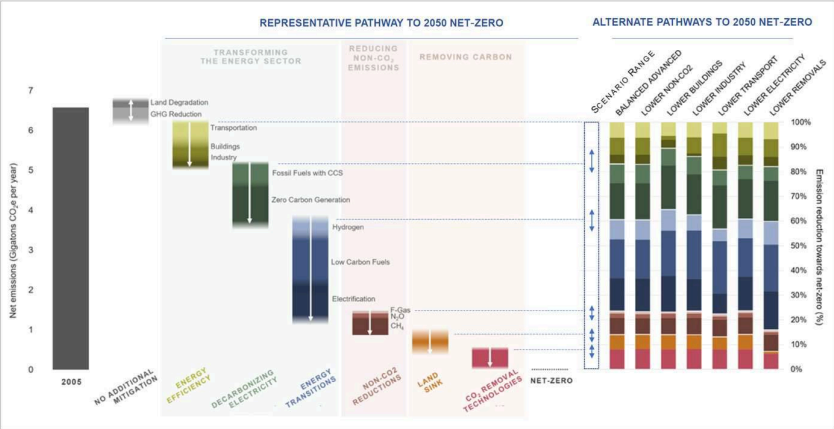


Figure 5: United States net zero carbon policies (United States Department of State and the United States Executive Office of the President, 2021, p. 23)

1.3 Project Details

Table 1: Project Details

Total Built-Up Area	3567 msq
Clients	• Pikes Peak – America's Mountain • United States Forest Service • Colorado Springs Utilities • United States Army • Broadmoor Manitou and Pikes Peak Cog Railway
Architect	RTA Architects GWWO Architects
Structural Engineer	HCDA Engineering
Civil Engineer	Kiowa Engineering Corporation
Mechanical Engineer/HVAC	JVA Engineering Inc.
Contractor	GE Johnson Construction Company
Permafrost Consultant	AECOM
Wind Engineer	CPP
Landscape Architect	DHM Design
Sustainability (LEED) Consultant	Ge Johnson
Start Of Construction	4th June 2018
Official Opening	30th June 2021
Certifications	LEED Silver Living Building Challenge V.3.1

Source: City of Colorado Springs (2016, p.3,4 and 5)

1.4 Pre-Design Post Occupancy Evaluation and Deconstruction of Old Summit House

At the summit of the mountain was an old visitors' house which had been in existence since 1837 which underwent a major renovation in 1963. Surveys were undertaken by the project team to gather information from citizens of Colorado Springs and 600,000 visitors that frequented the old summit house annually as seen in Figure 6. The old building was carefully

deconstructed simultaneously with the new building's construction and a few materials were salvaged and recycled.

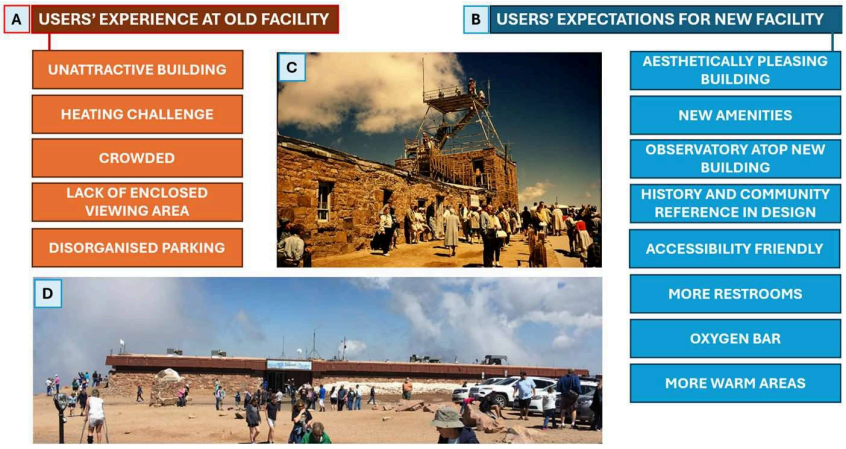


Figure 6: (A)Users' experience at old Summit building (B) Expectations for new facility (C) First Summit house built in 1873 (D) Renovated summit house in 1963 (City of Colorado Springs, n.d. and City of Colorado Springs 2017 p.11 and 15).

2.0 PASSIVE DESIGN STRATEGIES

2.1 Building site and orientation

Grable (2024) informs that, the building is oriented with the longest side facing south and south-east to maximize solar heat gain, daylighting, and energy conservation. Detailed wind studies and snowstorm analysis informed for the building to be partially buried into the south-east side of the mountain to shield it from the harsh winds and snowstorms.

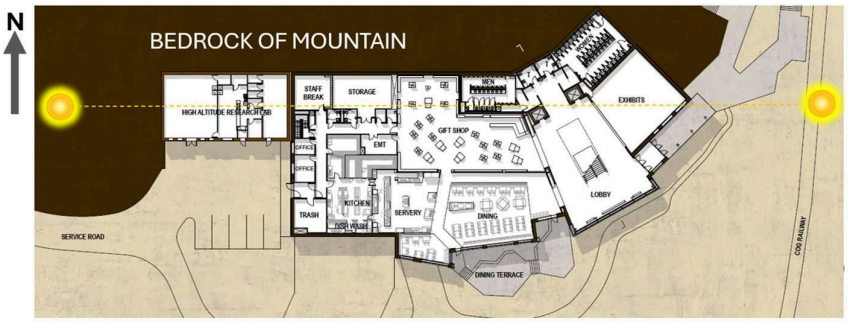


Figure 7: Floor plan of summit building (Colorado Springs 2017 p.34)

2.2 Shading

Colorado Springs (2016) describes the roof terrace at the southern end of the building cantilevering over the entrance walkway providing shade for the windows that lined along the façade. Sun breakers shade the windows that overlook the dining terrace. Other smaller windows employed on other facades were recessed and had shutters that protected them from storms.

2.3 Buffer zones

Colorado Springs (2016) further informs that, a small vestibule connects the main entrance doors to the double volume grand lobby with both spaces acting as a bridge between the extreme conditions outside and the warm indoor temperatures.

2.4 Daylighting

Even though the building has a wall-to-window ratio of 14 percent, natural lighting is a major consideration with adequate windows on all facades to usher in light to create dense light flooding into interior spaces Grable (2024).

2.5 Windows and glazing system

Dynamic glass was employed throughout the building forming part of the main south facing curtain wall system and other windows located on other facades. The glass had features that reflected out hot sun rays, reduced unwanted heat and glare from entering the spaces. Another layer of protection was given to the windows through the installation of metal shutters that would cover windows on facades at the windward side of the building against high storms (Grable, 2024).

2.6 Thermal Mass, Insulation and Air Tightness

Huso (2024) explains the conditions of the site required a super airtight and insulated building to prevent infiltration of exterior cold conditions. A high thermal mass consisting of precast concrete stored up heat and released into spaces at night. The best form of insulation from foundation up to the walls was Expanded Polystyrene (EPS) with a high resistance to water vapour to prevent heat loss and cold infiltration from the mountain.

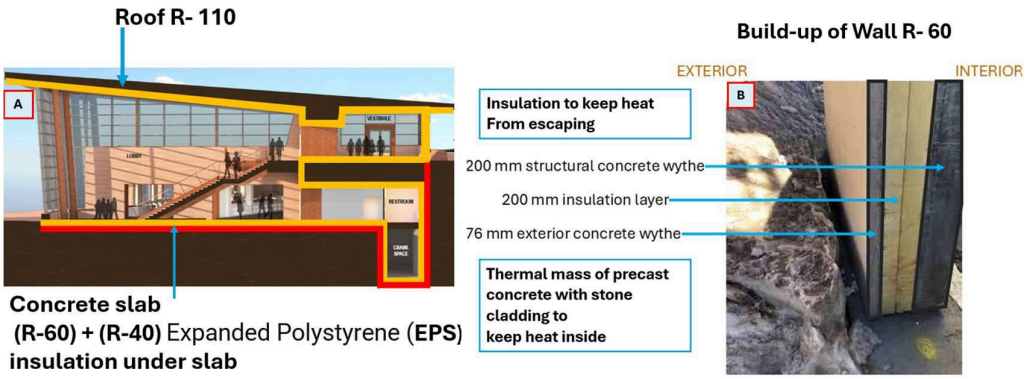


Figure 8:(A) Schematic section of building indicating layers of insulation (B) Expanded Polystyrene (EPS) sandwiched between two precast concrete walls (modified by Author; adapted from Colorado Springs (2017 p.36) and Huso (2024 p.36).

2.7 Envelope, Materials and Energy Implications

A well-insulated building envelope made up of a remarkably high U-value insulated precast concrete walls enclosed the entire building to provide shelter for guests indoors. The concrete and stone functioned as high thermal mass that stored up heat to be used in the spaces leading to reduced energy consumption (Huso 2024). Natural materials such as stone and wood were used for wall cladding and ceiling. All other materials were vetted to ensure they did not contain harmful chemicals and gases. Salvaged steel from old railway lines formed part of the building's structure and roof components (Grable, 2024).

3.0 BUILDING SERVICES AND ENERGY

3.1 Heating

According to Grable (2024) and City of Colorado Springs (2016), hydronic radiant floor system was the best option for heating up the entire building. The hot water tubes were laid below the concrete slabs which got heated up and the heat distributed to other spaces according to the heating zones of the building as seen in Fig. 9. From the entrance, the grand lobby had the highest temperature swings to aid the body temperature of guests to slowly adjust to the indoor temperature. The entire system is dependent on two energy efficient boilers for heating, circulation, and reheating.

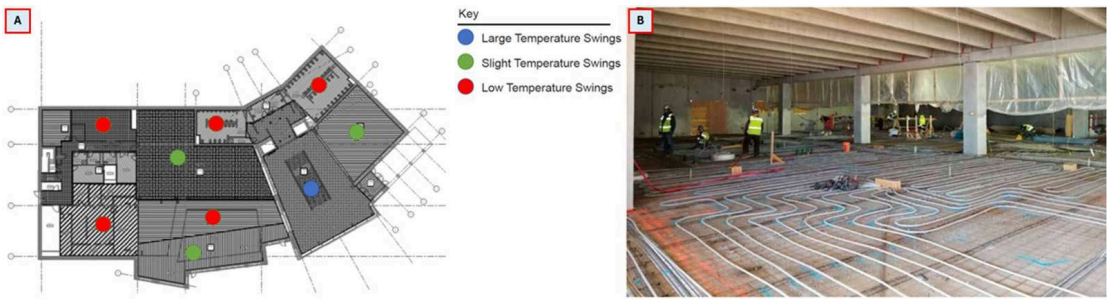


Figure 9: (A)Floor plan showing heating zones and temperature swings (B) Radiant floor system. (City of Colorado Springs (2016) and Grable (2024 p.102)

3.2 Mechanical Ventilation

Grable (2024) explains that, due to the extreme harsh freezing conditions, employing natural ventilation in the building was not expedient and this called for total reliance on mechanical ventilation with an energy recovery ventilator that harnessed the heat of outgoing warm air back into the interior spaces according to ASHRAE 62.2 requirements.

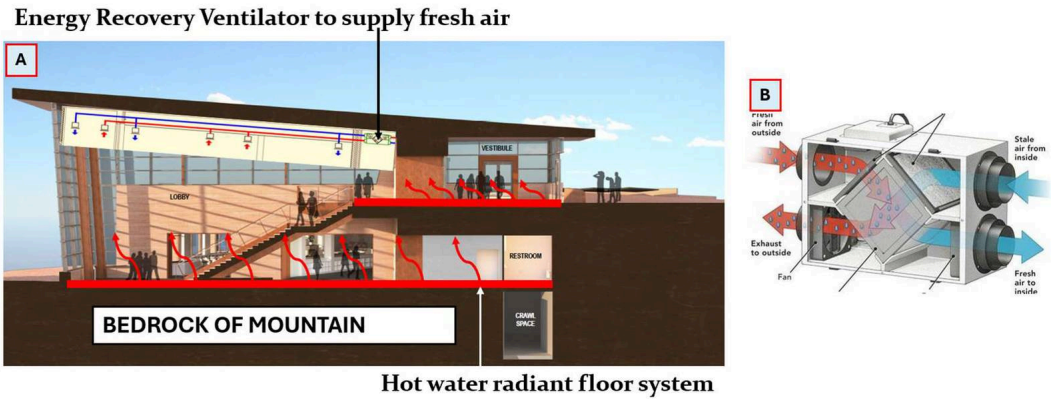


Figure 10: (A)Schematic section of building indicating radiant floor system (B) Energy recovery diagram (modified by author; adapted from Colorado Springs (2017 p.36) and Linquip Tech (2023)

3.3 Net Zero Water – Snow, Rainwater Harvesting and Water Treatment

Grable (2024) and Colorado Springs (2016) reported that water usage in the old summit building was high due to high water consumption facilities and lack of strict adherence to water saving. The new summit project incorporated vacuum flush toilets, waterless urinals, and low water use kitchen equipment. A step further was to harvest rain, snow and all wastewater, treat and re-use the water. Even though installation had been completed prior to handing over, it had not been running as at the early part of the year 2024.

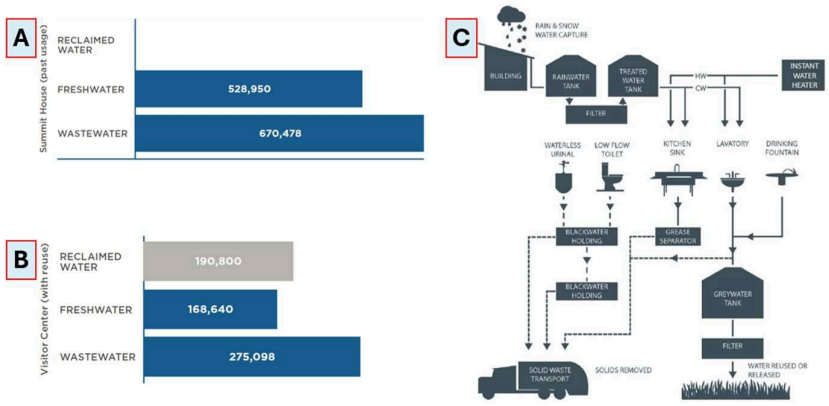


Figure 11: (A) Water usage of old summit building (B)Water usage of new summit building (C) Water harvesting, treatment, and re-use diagram (City of Colorado Springs (2016 p.28) and Grable (2024 p.79)

4.0 NET ZERO ENERGY STRATEGIES AND RENEWABLE ENERGY

Grable (2024) states that, various renewable options were explored by the design team and there was no possibility of incorporating a ground source heat pump due to the permafrost and cold temperatures within the mountain. Snowstorms and drift over the building wouldn't make it possible to install solar photovoltaic panels on the roof as seen in Figure 12 (A). The

Project invested in a 175-megawatt solar project jointly with Pikes Peak utility to produce adequate power for the summit building and also to power several homes within Colorado Springs as seen in Figure 12 (B). 630 kilowatts of the solar project powered the summit building based on its energy demand.

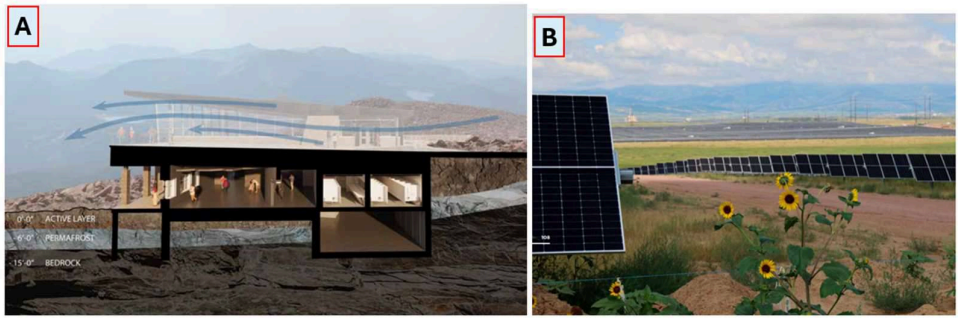


Figure 12: (A) Section indicating snowstorm pattern and cold temperatures within the mountain (B) 175-megawatt Solar Project located outskirts of Colorado Springs. (Grable 2024p.94 and 108)

4.1 Building Energy Performance

The previous summit building had a high energy consumption rate of 250-275 kbty/sf/yr and this formed the baseline for the new summit building, to create a better energy use and performance. With that as a base point, incorporating passive strategies and energy efficient active systems for heating, there was a huge energy reduction to 58 kbty/sf/yr (City of Colorado Springs, 2024) and (Grable, 2024). The solar power supplied to the building resulted in a net zero operational energy use as seen in Figure 13 (A) and (B).

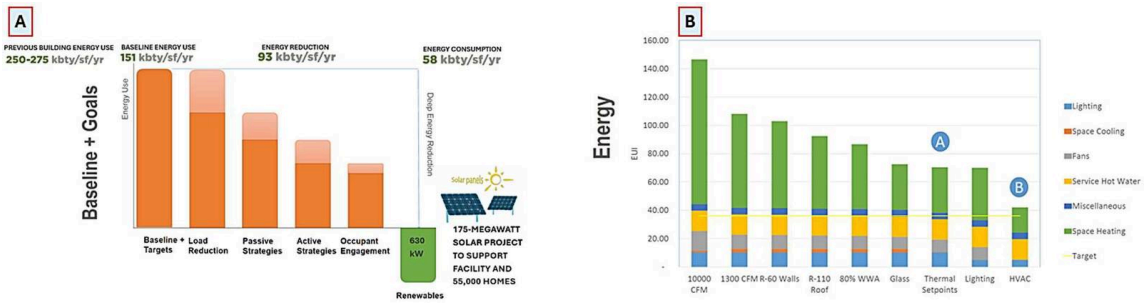


Figure 13: (A) Energy use baseline and actual consumption (B) Details of energy use (modified by author; adapted from City of Colorado Springs (2016 p.26 and 27)

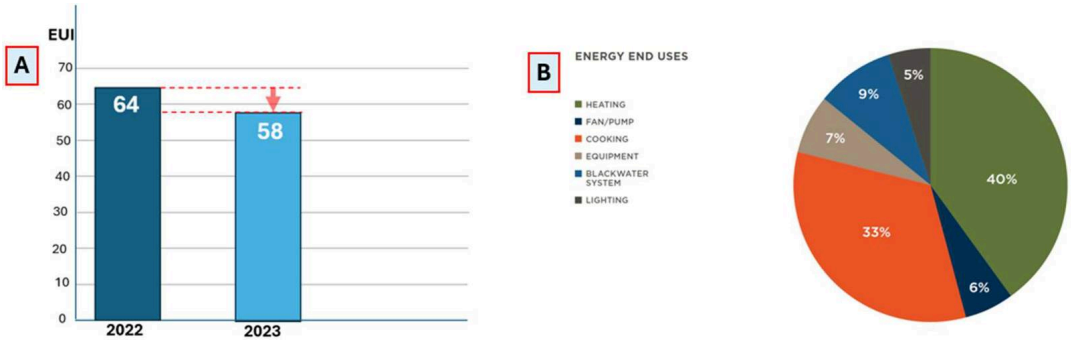


Figure 14: (A) Energy Use Intensity reduction (B) Energy end uses (Grable 2024 p.105 and106)

Grable (2024) explains that, even though all strategies had been in place to reduce the energy use, cooking and heating still stood as the major energy consumption units seen in Figure 14 (B). Incorporating heat recovery systems with the active strategies and introducing highly efficient kitchen equipment instead of conventional ones led to a huge reduction in the energy use intensity (EUI) as seen in Figure 14 (A).

5.0 LIFE CYCLE ASSESSMENT

The summit building has an extremely high embodied carbon due to the choice of materials for the construction; concrete, insulation and steel which were expedient to raise a robust building that could withstand the harsh conditions of the site that is, snow, wind and storm. As seen in Figure 15 (A and B), the product, that is, the building, has a high embodied carbon of 78 %. Transporting construction materials up the mountain significantly increased the total embodied carbon. Two materials with incredibly low embodied carbon that were incorporated in the project are natural stone and wood which gives a positive carbon score (Grable, 2024).

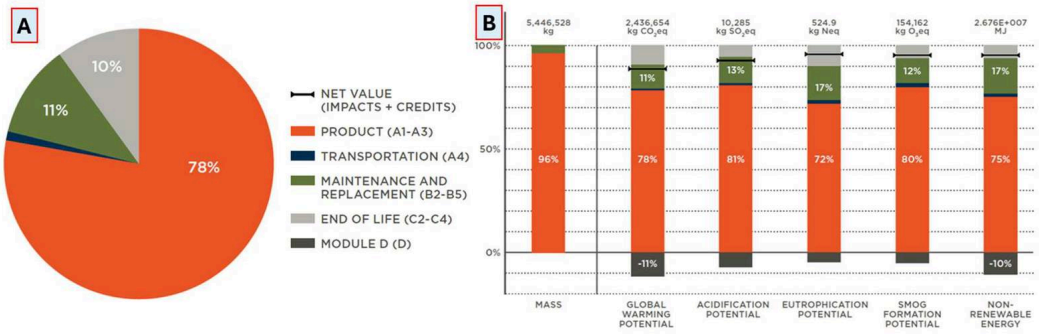


Figure 15: Life cycle assessment of Pikes Peak Summit building. Source: Grable (2024 p.172 and 173)

6.0 CERTIFICATIONS

Through the myriads of challenges during the design and construction, the project was awarded LEED silver the same year it was opened in 2021 after meeting all the requirements. It is still under assessment prior to being awarded the Living Building Challenge after meeting the criteria. The most challenging part in meeting both certifications was the materials components which had high embodied carbon (Grable, 2024).

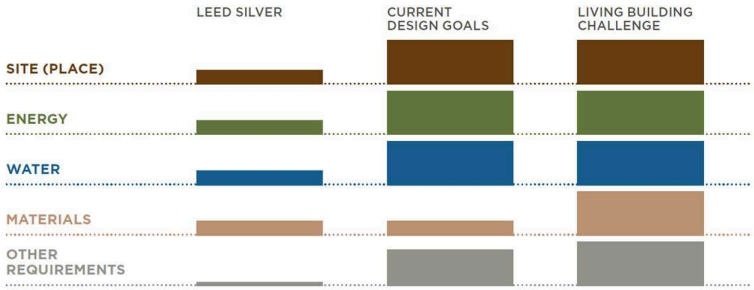


Figure 16: Certification goals. Grable (2024 p.36)

COMPARATIVE ANALYSIS AND CONCLUSION

The two projects have all strived to achieve net zero carbon emissions through design strategies and material choice.

Brock Environmental Centre located in a humid sub-tropical location incorporated several passive strategies including natural ventilation for most parts of the year which helps to reduce energy demand. Natural materials like wood and bio-based materials were the core materials used whiles achieving very high R-values. In meeting renewable energy strategies, solar panels, wind turbines and geothermal wells were the best considerations for the project achieving an Energy Use Intensity of 14.

Pikes Peak Visitor centre located at the top of a very harsh mountain also incorporates minimal passive strategies whiles heavily relying on mechanical ventilation, preventing infiltration of outdoor cold air into interior spaces. Extensive use of pre-cast concrete increased the embodied carbon of the building. It has a high Energy Use Intensity of 58 compared with Brock and its' renewable energy is solely from solar panels.

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